

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1966

West Memphis Gateway 100 kV
October 25, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1966. The Interconnection Customer proposes a 56.6 MW battery energy storage facility interconnecting at the West Memphis Gateway 100 kV substation in the Illinois area. The request was received on December 27, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 510 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$47,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Castle Afb / Shoreham Power Sta 100 KV line	Blackfoot / W Danville 15 230 KV	102.5	137	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Casle Afb / Shoreham Power Station 100 KV	Reconductor line	\$609,000	8% (\$47,000)
	Total allocated to project		\$47,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.